Brand Rule Lab

Product overview | CodeNexus India 2026 Round 3

Brand Rule Lab checks marketing content against approved brand rules, terminology, facts, evidence, assets, and claim history before publication. It helps teams find issues early, explains the evidence behind findings, and leaves publishing decisions with people.

Main workflow

1	A Creator submits a draft or supported visual asset.
2	The review engine runs deterministic checks and optional AI-assisted analysis.
3	Findings show the source, explanation, and suggested next step.
4	The Creator can propose a correction as a preserved new version.
5	An authorised Reviewer checks the current version and makes the approval decision.
6	Approvals and important workspace actions are recorded in an auditable history.

Implemented capabilities

Workspace and access

• Public landing page and sign-in; first-owner registration is available only before any account or workspace exists.

• Email/password account creation, email verification, password reset, invitation-only member registration, session revocation, and per-brand roles.

• Google OIDC can register the first owner or an invitee when the Google email is verified; existing verified accounts can link by matching email. Workspace membership remains the authorization boundary.

Content review

• Deterministic terminology, rule, claim, fact, evidence, and accessibility checks, with optional AI analysis after explicit consent.

• Claim remediation creates a new draft version and preserves the reason and finding traceability; a different Reviewer makes the final decision.

• Approved packaging references can be compared with uploaded package artwork using OCR token differences. This is an attention signal for a human reviewer.

Visual content and integrations

• Images support local OCR, exact SHA-256 identity, perceptual similarity, and approved asset references.

• MP4, MOV, and WebM uploads are bounded to 100 MB, 120 seconds, and up to 12 sampled keyframes by default. OCR text, frame timestamps, and extraction warnings are retained.

• The WordPress adapter creates and verifies draft posts only. Shopify and Contentful adapters remain local mocks; no adapter can publish.

History and evidence

• Claim snapshots retain links to the exact approved fact versions. Fact changes can flag claims for re-review.

• Append-only audit events form a verifiable hash chain with JSON export and a local PDF report.

Security and operating boundaries

• Passwords use scrypt; opaque local session tokens are stored as hashes, expire after eight hours, and can be revoked.

• Roles are checked in workflow services. Brand memberships scope workspace access, and the last Brand Owner cannot be removed.

• SQLite uses ordered additive migrations, foreign keys, WAL mode, and isolated test databases. Migration 14 adds email-verification and OIDC fields plus single-use token storage.

• Existing accounts with an email address but no verified-email timestamp must verify that address before email login, reset, OIDC linking, or email-bound invitations. Username login remains available for synthetic .invalid demo emails.

Limits and validation

• Video review checks sampled frames only; it does not verify unsampled moments, motion, disclaimer size, or logo safe zones. Packaging OCR can vary with image quality and does not compare layout, color, or visual placement.

• Google OAuth and Resend credentials were not available in this checkout. Their integrations have mocked tests but no live sign-in or delivery validation. Live WordPress was not tested against a real site.

• The full automated suite passed 323 tests with six PyMuPDF/SWIG deprecation warnings. Disposable v11-copy and fresh-database checks reached migration 14; all 73 baseline rows were preserved, SQLite integrity was ok, and foreign-key checks returned zero violations.

• The configured database was verified read-only at migration 13. No migration test used or modified that configured database.

• The local project and attachment search did not find the original Round 3 judging deck or rubric. The current requirement map uses only the Round 2 roadmap and is not the final judging checklist.

Local setup and deployment instructions are in README.md. Configure Resend with RESEND_API_KEY, EMAIL_FROM, and PUBLIC_APP_URL. Configure Google OIDC in Streamlit secrets using the app registered at /oauth2callback. Keep secrets out of source control.